

349. Intersection of Two Arrays:

Given two integer arrays `nums1` and `nums2`, return *an array of their intersection*. Each element in the result must be **unique** and you may return the result in **any order**.

Example 1:

```
Input: nums1 = [1,2,2,1], nums2 = [2,2]
Output: [2]
```

Example 2:

```
Input: nums1 = [4,9,5], nums2 = [9,4,9,8,4]
Output: [9,4]
Explanation: [4,9] is also accepted.
```

Constraints:

- `1 <= nums1.length, nums2.length <= 1000`
- `0 <= nums1[i], nums2[i] <= 1000`

Solution: time = $O(n*m)$ and space $O(\min(n,m))$

```
class Solution:
    def intersection(self, nums1: List[int], nums2: List[int]) -> List[int]:
        l, r = 0, 0
        res = []
        for i in nums1:
            if i in nums2 and i not in res:
                res.append(i)
```

```
return res
```

Set-method: Time=(m+n) and space = (m+n)

```
class Solution:
    def intersection(self, nums1: List[int], nums2: List[int]) -> List[int]:
        return list(set(nums1) & set(nums2))
```

Two-Pointer: time = $O(n \log n + m \log m)$ and space = $O(\min(n, m))$

```
class Solution:
    def intersection(self, nums1: List[int], nums2: List[int]) -> List[int]:

        nums1.sort()
        nums2.sort()

        l = 0
        r = 0
        res = []

        while l < len(nums1) and r < len(nums2):

            if nums1[l] == nums2[r]:

                if not res or res[-1] != nums1[l]:
                    res.append(nums1[l])

                l += 1
                r += 1

            elif nums1[l] < nums2[r]:
                l += 1
```

```
        else:  
            r += 1  
  
    return res
```
